Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/31**

Paper 3 Practical Test **May/June 2021**

CONFIDENTIAL INSTRUCTIONS

3825228465

This document gives details of how to prepare for and administer the practical exam.

The information in this document and the identity of any materials supplied by Cambridge International are confidential and must NOT reach candidates either directly or indirectly.

The supervisor must complete the report at the end of this document and return it with the scripts.

INSTRUCTIONS

- If you have any queries regarding these confidential instructions, contact Cambridge International stating the centre number, the syllabus and component number and the nature of the query.
  email info@cambridgeinternational.org
  phone +44 1223 553554

This document has **8** pages. Any blank pages are indicated.

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General information about practical exams

Centres must follow the guidance on science practical exams given in the *Cambridge Handbook*.

Safety

Supervisors must follow national and local regulations relating to safety and first aid.

Only those procedures described in the question paper should be attempted.

Supervisors must inform candidates that materials and apparatus used in the exam should be treated with caution. Suitable eye protection should be used where necessary.

The following hazard codes are used in these confidential instructions, where relevant:

C	corrosive	**MH**	moderate hazard
HH	health hazard	**T**	acutely toxic
F	flammable	**O**	oxidising
N	hazardous to the aquatic environment		

Hazard data sheets relating to substances used in this exam should be available from your chemical supplier.

Before the exam

- The packets containing the question papers must **not** be opened before the exam.
- It is assumed that standard school laboratory facilities, as indicated in the *Guide to Planning Practical Science*, will be available.
- Spare materials and apparatus for the tasks set must be available for candidates, if required.

During the exam

- It must be made clear to candidates at the start of the exam that they may request spare materials and apparatus for the tasks set.
- Where specified, the supervisor **must** perform the experiments and record the results as instructed. This must be done **out of sight** of the candidates, using the same materials and apparatus as the candidates.
- Any assistance provided to candidates must be recorded in the supervisor's report.
- If any materials or apparatus need to be replaced, for example, in the event of breakage or loss, this must be recorded in the supervisor's report.

After the exam

- The supervisor must complete a report for each practical session held and each laboratory used.
- Each packet of scripts returned to Cambridge International must contain the following items:
  - the scripts of the candidates specified on the bar code label provided
  - the supervisor's results relevant to these candidates
  - the supervisor's reports relevant to these candidates
  - seating plans for each practical session, referring to each candidate by candidate number
  - the attendance register.

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Specific information for this practical exam

During the exam, the supervisor (**not** the invigilator) must do all the experiments and record the results on a spare copy of the question paper, clearly labelled 'supervisor's results'.

If chemicals are prepared in more than one batch, clearly labelled supervisor's results must be provided for each batch. The candidates using each batch must be listed on the supervisor's report.

Apparatus

The apparatus listed must be provided to each candidate.

1 × 25 cm^3 pipette
1 × pipette filler
1 × 50 cm^3 burette
1 × stand
1 × burette clamp
1 × funnel for filling burette
1 × white tile
1 × conical flask suitable for titration
a supply of test-tubes
1 × test-tube rack (to support test-tubes and boiling tubes)
1 × test-tube holder (to hold test-tubes and boiling tubes)
1 × stirring rod
2 × boiling tubes
1 × Bunsen burner
1 × heat-proof mat
4 × teat/dropping pipettes
1 × beaker (for washing teat/dropping pipettes)
1 × spatula
1 × wash bottle containing distilled water
paper towels
red and blue litmus papers or universal indicator paper
wooden splints
apparatus normally used in the centre in testing for carbon dioxide with limewater

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Materials

The materials listed in the table must be provided to each candidate.

Warning: small amounts of NH_3 **[C] [T] [N]**, which can cause respiratory distress in some people, may be produced. **The laboratory must be well ventilated.**

label		per candidate	identity	notes
	P	150 cm^3	0.04 mol / dm^3 dilute sulfuric acid	Cautiously pour 2.2 cm^3 of concentrated (98%) sulfuric acid **[C]** into 500 cm^3 of distilled water with continuous stirring. Make the solution up to 1 dm^3 with distilled water.
[MH]	**Q**	150 cm^3	0.10 mol / dm^3 potassium hydroxide solution	Dissolve 5.611 g of KOH **[C] [MH]** in each dm^3 of solution.
[F] [MH] [HH]	thymolphthalein indicator	5 cm^3	thymolphthalein indicator	Dissolve 2.0 g in 1 dm^3 of industrial denatured alcohol (IDA) **[F] [MH] [HH]**.
Supervisors are asked to carry out a standard acid/base titration between solutions **P** and **Q** to ensure that the concentrations of the two solutions fall within the given range. It is essential that 25.0 cm^3 of **Q** reacts with between 29.0 cm^3 and 33.0 cm^3 of **P**.				
	R	10 cm^3	0.1 mol / dm^3 $CrCl_3$	Dissolve 26.7 g of $CrCl_3 \cdot 6H_2O$ in 700 cm^3 of cold distilled water and make up to 1000 cm^3 with distilled water.
	S	10 cm^3	0.4 mol / dm^3 iron(II) sulfate solution	Dissolve 111.20 g of $FeSO_4 \cdot 7H_2O$ in 700 cm^3 of cold distilled water and make up to 1000 cm^3 with distilled water. This solution must be freshly prepared.
[MH]	aqueous hydrogen peroxide	10 cm^3	20 vol H_2O_2	Dilute 200 cm^3 of 100 vol (approximately 8.3 mol / dm^3) H_2O_2 **[C]** with 700 cm^3 of distilled water and make up to 1000 cm^3. This solution must be freshly prepared using **newly purchased** hydrogen peroxide.

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label		per candidate	identity	notes
[C]	dilute nitric acid	10 cm^3	1.0 mol / dm^3 HNO_3	See preparation instructions on page 30 of the 2020–2021 syllabus. If necessary, each of these reagents can be provided as a communal supply for groups of up to 6 candidates. Invigilators must be alert to the risk of contamination and the opportunity for malpractice when using a communal supply.
[MH] [N]	aqueous ammonia	10 cm^3	1.0 mol / dm^3 NH_3	
[C]	aqueous sodium hydroxide	10 cm^3	1.0 mol / dm^3 NaOH	
	aqueous silver nitrate	10 cm^3	0.05 mol / dm^3 $AgNO_3$	
	aqueous barium nitrate	10 cm^3	0.1 mol / dm^3 $Ba(NO_3)_2$ **or** $BaCl_2$	
[MH]	limewater	10 cm^3	saturated aqueous calcium hydroxide, $Ca(OH)_2$	

- An excess of at least 10% of each material must be prepared to cover accidental loss.
- All solutions must be thoroughly mixed.
- If you are unable to source any of these chemicals, you must contact Cambridge International as far as possible in advance of the exam for advice.
- **Materials must be labelled only as specified in the 'label' column. The identities of chemicals labelled with letter codes, e.g. P, may be different from their descriptions in the question paper. Candidates must use the descriptions given in the question paper.**

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Supervisor's report

Syllabus and component number | | | | | / | | |

Centre number | | | | | |

Centre name .....

Time of the practical session .....

Laboratory name/number .....

Give details of any difficulties experienced by the centre or by candidates (include the relevant candidate names and candidate numbers).

You must include:

- any difficulties experienced by the centre in the preparation of materials
- any difficulties experienced by candidates, e.g. due to faulty materials or apparatus
- any specific assistance given to candidates.

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If chemicals have been prepared in more than one batch, list the candidates using each batch.

Declaration

1 Each packet that I am returning to Cambridge International contains all of the following items:
- the scripts of the candidates specified on the bar code label provided
- the supervisor's results relevant to these candidates
- the supervisor's reports relevant to these candidates
- seating plans for each practical session, referring to each candidate by candidate number
- the attendance register.

2 Where the practical exam has taken place in more than one practical session, I have clearly labelled the supervisor's results, supervisor's reports and seating plans with the time and laboratory name/number for each practical session.

3 I have included details of difficulties relating to each practical session experienced by the centre or by candidates.

4 I have reported any other adverse circumstances affecting candidates, e.g. illness, bereavement or temporary injury, directly to Cambridge International on a *special consideration form*.

Signed ..... (supervisor)

Name (in block capitals) .....

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